

MODULE 3: PROGRAMMING FOR BIOLOGICAL DATA

Hypothesis Testing in the Clinical Laboratory

Session 5: T-tests, Normality, and P-values

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Learning Objectives



Distribution

Understand the Normal Distribution and its critical role in reference intervals.



Normality

Test data for normality using `shapiro.test()` and Q-Q plots.



T-Tests

Perform one-sample, paired, and two-sample t-tests in R.



Interpretation

Interpret p-values correctly in clinical decision-making contexts.

Why Hypothesis Testing?

The Questions We Ask

- ❓ Is this new reagent lot equivalent to the old one?
- ❓ Is our analyzer calibrated correctly?
- ❓ Do viral loads differ between patient groups?

The Objective Answer

Hypothesis testing provides a rigorous statistical framework to answer these questions objectively, moving beyond "it looks different" to "it is statistically different."

Part 1: The Normal Distribution

Foundations of Clinical Statistics

Duration: 40 minutes

Why "Normal" Matters in the Lab

The **Normal (Gaussian) Distribution** is the foundation of many laboratory practices.

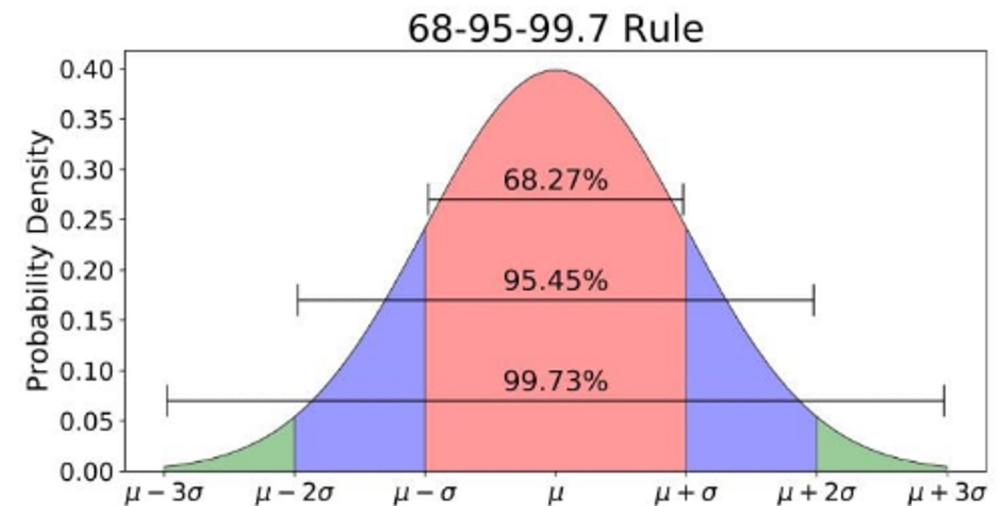
Application	How Normal Distribution is Used
Reference Intervals	The standard "Mean $\pm$ 2SD" calculation covers 95% of a healthy population only if the data is normally distributed.
Quality Control	Levey-Jennings charts rely on the assumption that control values follow a normal distribution.
Statistical Tests	Most parametric tests (like t-tests and ANOVA) assume normality to be valid.

The Bell Curve

Visualizing Hemoglobin

Imagine measuring hemoglobin in 1000 healthy adults:

- Most values **cluster around the center** (the mean).
- Fewer values exist at the **extremes** (very low or very high).
- The shape is perfectly **symmetric**.



Key Properties

Standard Deviation Rules

- **Mean $\pm$ 1SD:** Covers ~68% of values
- **Mean $\pm$ 2SD:** Covers ~95% of values
- **Mean $\pm$ 3SD:** Covers ~99.7% of values

Clinical Application

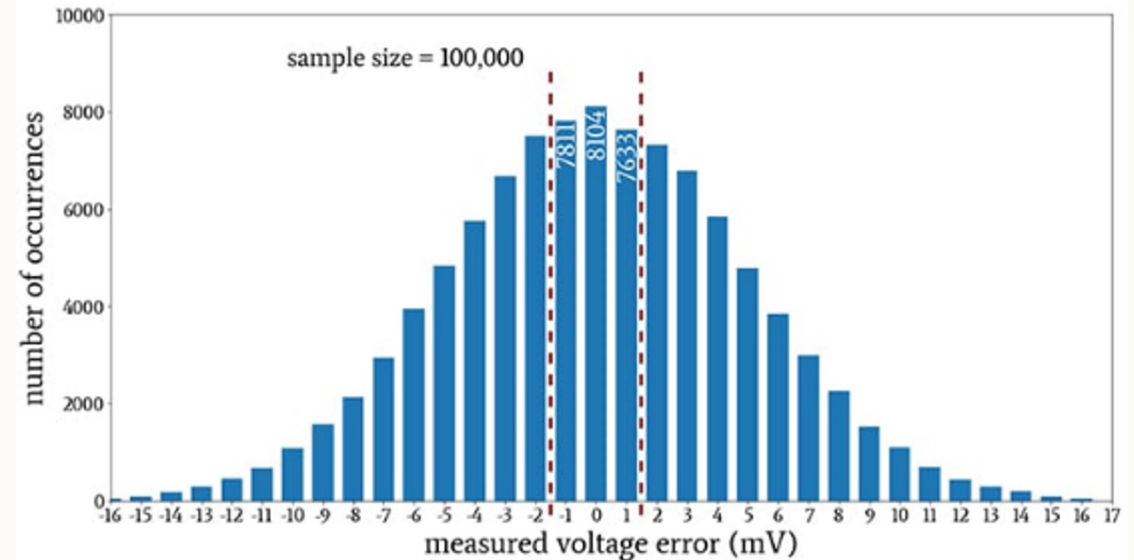
Reference Interval $\approx$ Mean $\pm$ 2SD

This captures the central 95% of the healthy population.

Visualizing the Normal Distribution in R

```
# Simulating hemoglobin values
set.seed(123)
hemoglobin <- rnorm(n = 1000, mean = 14.0, sd = 1.5)

# Histogram with reference lines
hist(hemoglobin, breaks = 30, col = "steelblue")
abline(v = mean(hemoglobin), col = "red", lwd = 2)
# Add +/- 2SD lines
sd_val <- sd(hemoglobin)
abline(v = mean(hemoglobin) + 2*sd_val,
      col = "orange", lty = 2)
abline(v = mean(hemoglobin) - 2*sd_val,
      col = "orange", lty = 2)
```



Calculating Reference Intervals

R Code

```
mean_hb ← mean(hemoglobin)
sd_hb ← sd(hemoglobin)

lower_ref ← mean_hb - 2 * sd_hb
upper_ref ← mean_hb + 2 * sd_hb

cat("Ref Interval:", lower_ref, "-",
    upper_ref)
```

Expected Output

Based on our simulated data (Mean=14, SD=1.5):

Reference Interval: 11.0 - 17.0 g/dL

This interval would theoretically contain 95% of healthy individuals.

Testing for Normality

Before using parametric tests (like t-tests), you must verify your data is approximately normal.

Method	Type	Tool in R	What to look for
Q-Q Plot	Visual	qqnorm(), qqline()	Points falling along the diagonal line.
Shapiro-Wilk	Statistical	shapiro.test()	P-value > 0.05 (Fail to reject null).

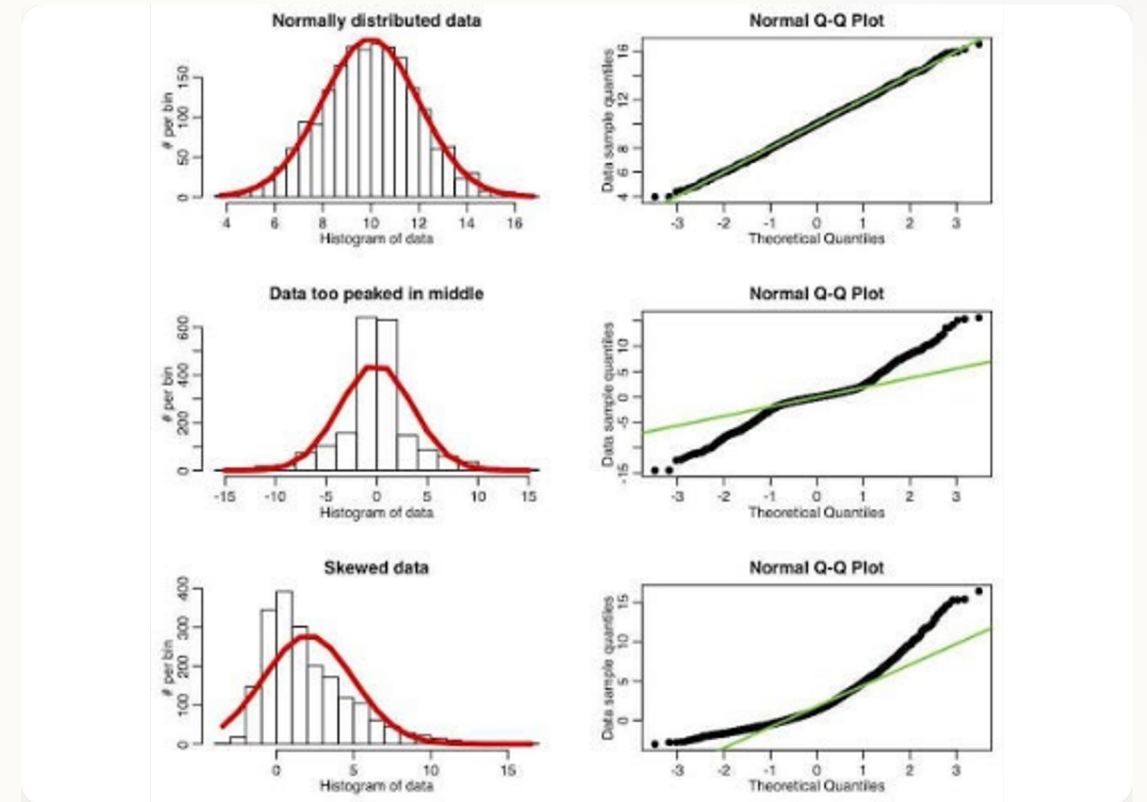
Q-Q Plot: Visual Normality Check

How to Read It

A Q-Q (Quantile-Quantile) plot compares your data against a theoretical normal distribution.

✓ **Normal:** Points follow the straight diagonal line closely.

⚠ **Skewed/Non-normal:** Points curve away from the line (S-shape or banana shape).



Live Demo: Q-Q Plots in R

Code

```
par(mfrow = c(1, 2))

# Normal data (Hemoglobin)
qqnorm(hemoglobin, main = "Normal Data")
qqline(hemoglobin, col = "red")

# Skewed data (Viral loads)
viral_loads <- rexp(100, rate = 0.001)
qqnorm(viral_loads, main = "Skewed Data")
qqline(viral_loads, col = "red")
```

Visual Inspection

The code on the left generates two plots side-by-side.

Notice how the **Normal Data** hugs the red line, while the **Skewed Data** (exponential distribution) deviates significantly at the ends.

Shapiro-Wilk Test

The Hypothesis

- H_0 (Null): Data is normally distributed.
- H_1 (Alternative): Data is NOT normally distributed.

Interpretation

```
shapiro.test(hemoglobin)  
# p-value = 0.72
```

$p \geq 0.05$: Fail to reject $H_0 \rightarrow$ Data may be normal.

$p < 0.05$: Reject $H_0 \rightarrow$ Data is NOT normal.

Clinical Relevance: CLSI C28-A3

1

Test First

Always test reference population data for normality before calculating intervals.

2

Non-Parametric

If data is **not normal** (and cannot be transformed), use non-parametric methods (percentiles) instead of $\text{Mean} \pm 2\text{SD}$.

3

Transform

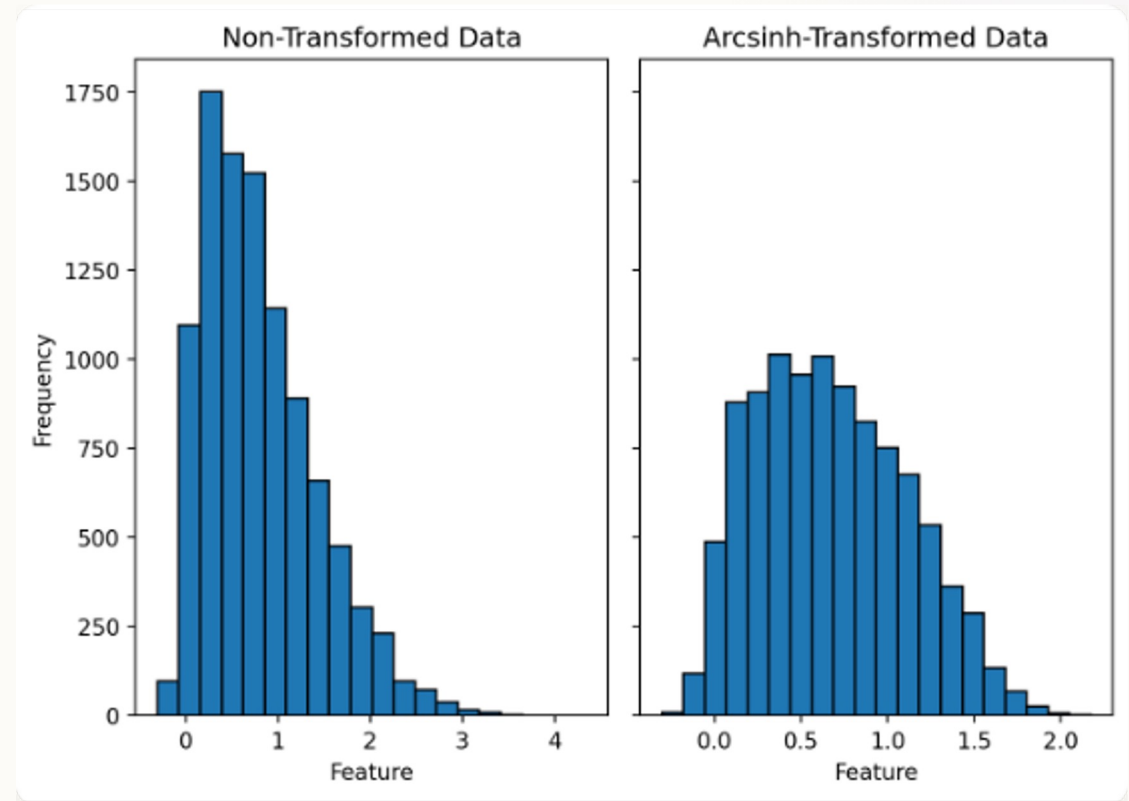
Consider data transformation (e.g., Log10) to normalize skewed distributions like viral loads.

Log Transformation for Skewed Data

Viral loads are typically right-skewed. Log-transformation often normalizes them, allowing the use of parametric tests.

```
log_viral ← log10(viral_loads + 1)

# Check normality after transform
shapiro.test(log_viral)
# Result: p > 0.05 ✓
```



Part 2: T-Tests — Comparing Means

Statistical Comparisons in the Lab

Duration: 40 minutes

The Logic: A Courtroom Analogy

Courtroom Concept	Hypothesis Testing Equivalent
Innocent until proven guilty	H_0 : No difference (Null Hypothesis is assumed true initially)
Evidence must be convincing	P-value measures the strength of evidence against H_0
"Beyond reasonable doubt"	$p < 0.05 \rightarrow$ Reject H_0 (Statistically Significant)
"Not enough evidence" $\neq$ Innocent	$p \geq 0.05 \rightarrow$ Fail to reject H_0 (Not proven equal, just no difference found)

Understanding the P-value

"The probability of obtaining results at least as extreme as the observed results,
assuming that the null hypothesis is correct."

$$p < 0.05$$

Statistically Significant

Strong evidence against H_0 .

$$p \geq 0.05$$

Not Significant

Insufficient evidence to reject H_0 .

Which T-Test to Use?

Scenario	Test Type	R Function
Compare a sample mean to a known standard/target .	One-sample t-test	<code>t.test(x, mu = value)</code>
Compare same subjects under two conditions (e.g., Method A vs B).	Paired t-test	<code>t.test(x, y, paired = TRUE)</code>
Compare two independent groups (e.g., Delta vs Omicron patients).	Two-sample t-test	<code>t.test(x, y)</code>

Scenario 1: One-Sample T-Test

Clinical Scenario

You receive a **new lot of QC material**.

- Manufacturer states target glucose = **100 mg/dL**.
- You run **20 replicates** on your analyzer.

The Question

Does your analyzer measure at the target?

H_0 : Mean = 100 mg/dL

H_1 : Mean $\neq$ 100 mg/dL

One-Sample T-Test in R

```
qc_glucose <- c(101, 103, 99, 102, 100, 104, 98, 103, 101, 102, 100, 105, 99,  
               101, 103, 102, 100, 104, 101, 103) target <- 100 # Perform t-test  
t.test(qc_glucose, mu = target) # Output: # t = 3.25, df = 19, p-value = 0.004  
# mean of x: 101.55 # 95% CI: [100.5, 102.3]
```

Interpreting the Output

P-value

0.004

Since $p < 0.05$, we reject H_0 . There is a statistically significant difference.

Mean Estimate

101.55

Our measured mean is ~1.55 mg/dL higher than the target of 100.

Decision

Bias Detected

The analyzer has a significant positive bias.

Scenario 2: Paired T-Test (Method Comparison)

The Setup

Validating a **new reagent (Method B)** against the current one (Method A).

- Measure **20 patients** with BOTH methods.
- Samples are **paired** because measurements come from the same patient.

Why "Paired"?

We analyze the differences (A - B) for each patient.

If Mean Difference $\approx 0 \rightarrow$ Methods are equivalent.

If Mean Difference $\neq 0 \rightarrow$ Systematic bias exists.

Paired T-Test in R

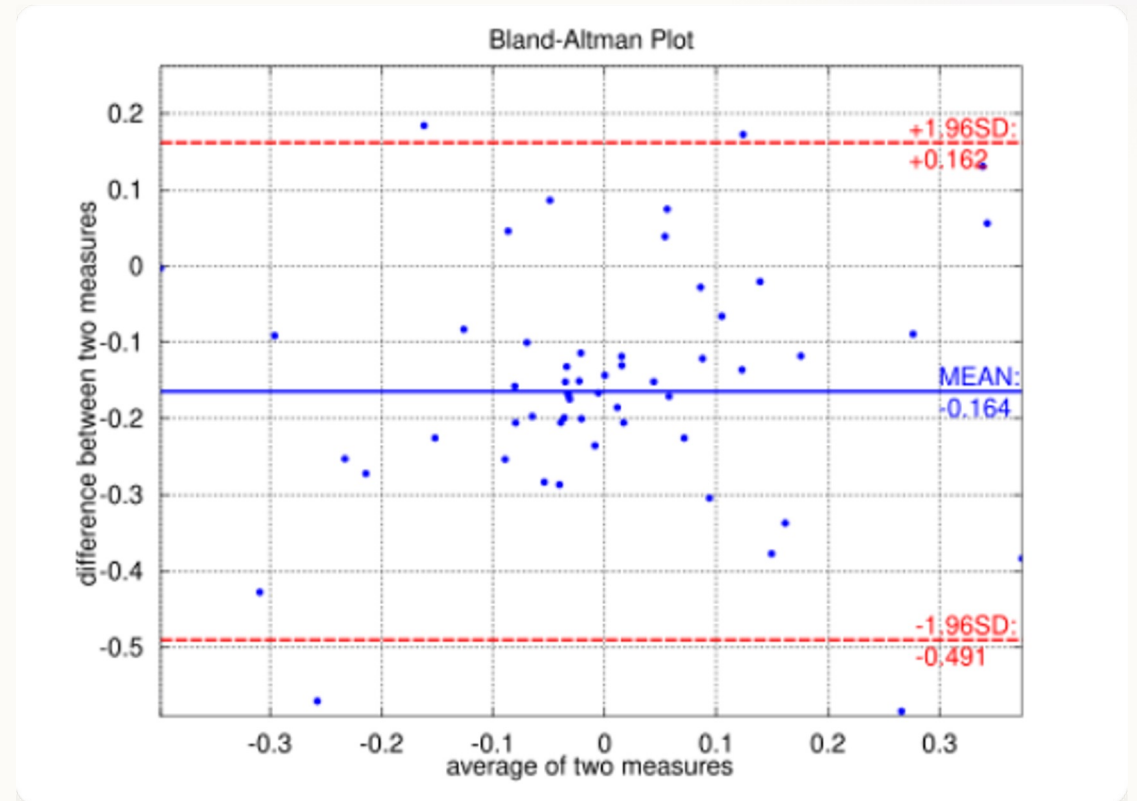
```
# Check normality of differences first!  
differences <- method_A - method_B  
shapiro.test(differences) # p > 0.05 ✓  
  
# Paired t-test  
t.test(method_A, method_B, paired = TRUE)  
  
# Output:  
# p-value = 0.001  
# mean of the differences: 1.5
```

Result: A statistically significant bias of 1.5 mg/dL exists between the methods.

Visualizing Difference: Bland-Altman Plot

The **Bland-Altman (Difference) plot** is the standard for method comparison.

- **X-axis:** Mean of the two methods $((A+B)/2)$.
- **Y-axis:** Difference between methods $(A-B)$.
- **Horizontal Lines:** Mean Bias (solid) and 95% Limits of Agreement (dashed).



Statistical vs. Clinical Significance

Critical Concept!

Even if $p < 0.05$ (statistically significant), you must ask:

"Is the difference clinically meaningful?"

A 1.5 mg/dL bias in glucose might be statistically significant but is likely **clinically irrelevant** given biological variation.



Making a Clinical Decision

The Logic

Compare the observed bias against **Total Allowable Error (TAE)** specifications.

For Glucose, TAE is typically ± 6 mg/dL or 10%.

R Implementation

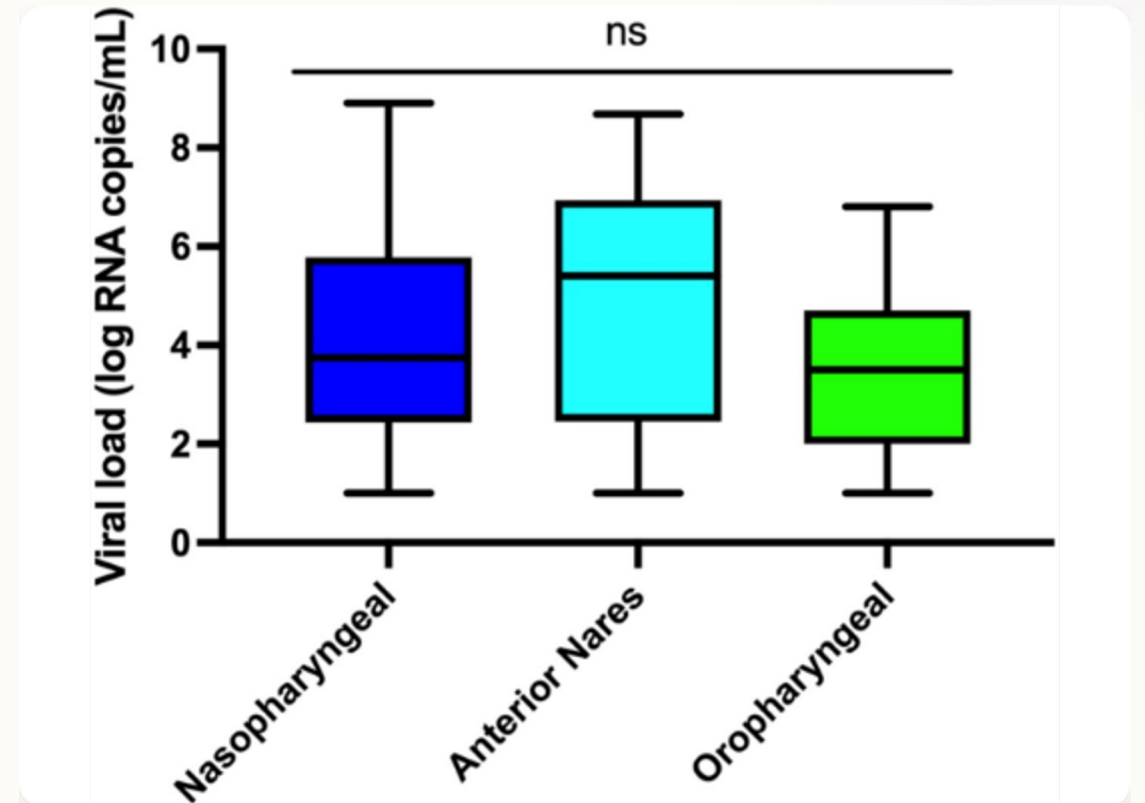
```
observed_bias ← abs(mean(differences))  
# 1.5 mg/dL  
  
clinical_threshold ← 6 # Glucose TAE  
  
if (observed_bias ≤ clinical_threshold) {  
  cat("✅ Clinically acceptable")  
} else {  
  cat("⚠️ Bias exceeds clinical limits")  
}
```

Scenario 3: Two-Sample (Independent) T-Test

Clinical Scenario

Do COVID-19 patients infected with **Delta** variant have different viral loads than **Omicron**?

- Two **independent** groups of patients.
- Different sample sizes possible.



Interpreting Variant Comparison

```
t.test(delta_ct, omicron_ct)  
# p-value < 0.001
```

Results

- Delta Mean Ct: 18.6
- Omicron Mean Ct: 21.8
- Difference: 3.2 cycles

Interpretation

Recall that PCR Ct values are logarithmic.

A difference of ~3.3 Ct represents a **10-fold difference** in viral copies.

Conclusion: Delta patients had significantly higher viral loads (~10x).

Before ANY T-Test: Check Assumptions

1

Normality

Use Shapiro-Wilk or Q-Q plots. If failed, use **Wilcoxon** (non-parametric).

2

Independence

Samples must be independent (or properly paired for paired tests).

3

Variance

For two-sample tests, check equal variance. If unequal, use **Welch's t-test** (default in R).

Summary: Which T-Test?

Scenario	Test	Key Assumption
Sample vs. Known Value	One-sample	Sample is normal
Same subjects, 2 conditions	Paired	Differences are normal
Two independent groups	Two-sample	Both groups normal

Always: Check normality first!

Tutorial 5

Method Validation Study: Practical
Application

Open your R environment

Tutorial Scenario

Role

You are a **Medical Laboratory Technologist** validating a new glucose reagent lot.

Task

- **Data:** 20 patient samples measured by Method A (Current) and Method B (New).
- **Goal:** Determine if there is systematic bias.
- **Question:** "Is the new lot statistically and clinically equivalent?"

What You'll Practice

Steps 1-2

Load Data & Calculate
Differences

Step 3

Test Normality
(Histogram/Shapiro)

Step 4

Perform Paired T-Test

Step 5-6

Clinical Significance &
Bland-Altman Plot

Step 1 & 2: Load and Difference

Load Data

```
data <- data.frame(  
  ID = paste0("P", 1:20),  
  Method_A = c(95, 120, ... ),  
  Method_B = c(97, 118, ... )  
)
```

Calculate Differences

```
data$Diff <- data$Method_A - data$Method_B  
  
mean_diff <- mean(data$Diff)  
sd_diff <- sd(data$Diff)  
  
# Think: What does a negative mean indicate?
```

Step 3: Test Normality

Before the t-test, verify the **differences** are normal.

```
# 1. Histogram  
hist(data$Diff)  
  
# 2. Shapiro-Wilk  
shapiro.test(data$Diff)
```

Rule: If $p \geq 0.05$, the data is normal, and we can proceed with the paired t-test.

Step 4: Perform Paired T-Test

```
result <- t.test(data$Method_A, data$Method_B, paired = TRUE)  
print(result)
```

Find these values in output:

- **t statistic:** Magnitude of difference relative to variance.
- **p-value:** Significance of the difference.
- **95% CI:** The range of the true bias.
- **mean of differences:** The observed bias.

Step 5 & 6: Significance & Plot

Check Clinical Limits

```
bias ← abs(mean(data$Diff))  
limit ← 6  
  
if (bias ≤ limit) {  
  print("✅ Acceptable")  
}
```

Bland-Altman Plot

```
plot(rowMeans(data[,2:3]), data$Diff)  
abline(h = mean(data$Diff), col="red")  
# Add +/- 1.96 SD lines
```

Challenge: Validation Report

Write a 3-5 sentence summary including:

- 1 Number of samples compared (n=20).
- 2 Mean bias and direction (Method A vs B).
- 3 Statistical significance (p-value).
- 4 Clinical acceptability (vs. ± 6 mg/dL).
- 5 **Final Recommendation:** Approve or Reject?